Supplementary Material: Limits to improving reproduction number estimates by increasing case counts: Environmental variability, reporting heterogeneity, and surveillance design

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S1. Mathematical proofs and analytical derivations

This section provides complete mathematical proofs for all propositions, corollaries, and remarks in the main text, alongside detailed derivations of the renewal process variance decompositions and higher-order delta expansions. Equation and theorem numbering from the main text is preserved; equations introduced solely in this supplement carry the prefix S.

Recall that the negative-binomial distribution $\mathrm{NB}(r, p)$ parameterized by number of failures r and success probability p has mean $r(1-p)/p$ and variance $r(1-p)/p^2$. Setting $r = k$ and $p = k/(k+R)$ yields Eq. (1). Convolutions and binomial thinning follow directly from infinite divisibility and thinning properties of the negative-binomial family (Casella and Berger, 2002).

S1.1. Proof of Proposition 1 (Estimation of Target 1)

Let c denote an observed realization of C_t, and let $\theta = \rho R_t$. From Eq. (3), the conditional log-likelihood of R_t given n is

$$\ell(R_t) = \text{const} + nk \log\left(\frac{k}{k+\theta}\right) + c \log\left(\frac{\theta}{k+\theta}\right). \tag{S1}$$

Differentiating with respect to R_t, the score function is

$$\dot{\ell}(R_t) = \frac{c}{R_t} - \frac{(c+nk)\rho}{k+\rho R_t} = \mathcal{I}(R_t)\,(\widehat{R}_t - R_t), \qquad \mathcal{I}(R_t) = \frac{k\rho n}{R_t(k+\rho R_t)} = \frac{n}{R_t/\rho + R_t^2/k}. \tag{S2}$$

(i) When $c > 0$, Eq. (S2) implies $\dot{\ell}(R_t) > 0$ for $R_t < \widehat{R}_t$ and $\dot{\ell}(R_t) < 0$ for $R_t > \widehat{R}_t$. Hence, the log-likelihood is strictly concave and achieves its unique global maximum at $\widehat{R}_t = c/(\rho n)$. When $c = 0$, $\dot{\ell}(R_t) = -nk\rho/(k + \rho R_t) < 0$ for all $R_t > 0$, so the maximum is attained at the boundary $R_t = 0 = \widehat{R}_t$.

(ii) Conditional expectation is $\mathbb{E}(\widehat{R}_t \mid R_t) = \rho n R_t/(\rho n) = R_t$, proving unbiasedness. Conditional variance is

$$\mathrm{Var}(\widehat{R}_t \mid R_t) = \frac{\rho n R_t + \rho^2 n R_t^2/k}{(\rho n)^2} = \frac{R_t}{\rho n} + \frac{R_t^2}{kn} = \frac{R_t(k + \rho R_t)}{k\rho n}. \tag{S3}$$

(iii) The Fisher information is the variance of the score function:

$$\mathcal{I}(R_t) = \mathrm{Var}(\dot{\ell}(R_t)) = \frac{(k\rho n)^2}{[R_t(k + \rho R_t)]^2}\mathrm{Var}(\widehat{R}_t) = \frac{k\rho n}{R_t(k + \rho R_t)} = \frac{n}{R_t/\rho + R_t^2/k}. \tag{S4}$$

Thus, $\mathrm{Var}(\widehat{R}_t \mid R_t) = 1/\mathcal{I}(R_t)$. Because the score function is strictly proportional to $\widehat{R}_t - R_t$, equality in the Cramér–Rao inequality is attained for all finite n (Casella and Berger, 2002).

(iv) Writing the probability mass function of Eq. (3) in natural exponential family form, the natural parameter is $\upsilon = \log[\rho R_t/(k + \rho R_t)]$. As $R_t \in (0, \infty)$, υ spans $(-\infty, 0)$, which contains an open interval. Therefore, C_t is a complete sufficient statistic. Because $\widehat{R}_t$ is an unbiased function of C_t, by the Lehmann–Scheffé theorem it is the unique uniformly minimum-variance unbiased estimator (UMVUE) (Lehmann and Romano, 2005). □

S1.2. Proof of Proposition 2 (Estimation of Target 2)

Under H4, $\mathbb{E}(R_t \mid n) = \bar{R}$ and $\mathrm{Var}(R_t \mid n) = v_R$. By Proposition 1, $\mathbb{E}(\widehat{R}_t \mid n, R_t) = R_t$, so $\mathbb{E}(\widehat{R}_t \mid n) = \bar{R}$. By the law of total variance:

$$\begin{aligned}
\mathrm{Var}(\widehat{R}_t \mid n) &= \mathbb{E}\big[\mathrm{Var}(\widehat{R}_t \mid n, R_t) \mid n\big] + \mathrm{Var}\big[\mathbb{E}(\widehat{R}_t \mid n, R_t) \mid n\big] \\
&= \mathbb{E}\left[\frac{R_t}{\rho n} + \frac{R_t^2}{kn} \;\middle|\; n\right] + \mathrm{Var}(R_t \mid n) \\
&= \frac{\bar{R}}{\rho n} + \frac{\bar{R}^2 + v_R}{kn} + v_R.
\end{aligned} \tag{S5}$$

Substituting $m = \rho n$ and collecting terms in $1/m$ yields Eq. (6).

(i) As $m \to \infty$, $A/m \to 0$, hence $V(m) \to v_R$.

(ii) The sampling fraction of total variance is $(A/m)/(v_R + A/m) = 1/(1 + mv_R/A) = 1/(1 + m/m_\times)$.

(iii) Proportional variance reduction upon doubling sample size is $[V(m) - V(2m)]/V(m) = [A/(2m)]/(v_R + A/m) = 1/[2(1 + m/m_\times)]$.

(iv) $V'(m) = -A/m^2 < 0$ and $V''(m) = 2A/m^3 > 0$, proving strict monotonicity and convexity. The scale elasticity is $\epsilon_m = mV'(m)/V(m) = -(A/m)/(v_R + A/m) = -m_\times/(m + m_\times)$, which evaluates to $-1/2$ at $m = m_\times$.

(v) Let $\tilde{\theta}$ be any estimator based on current-period data satisfying $\mathbb{E}(\tilde{\theta} \mid R_t) = R_t$. By the Cramér–Rao inequality, $\mathrm{Var}(\tilde{\theta} \mid R_t) \geq 1/\mathcal{I}(R_t)$. Substituting into the orthogonal decomposition $\mathbb{E}(\tilde{\theta} - \bar{R})^2 = \mathbb{E}[\mathrm{Var}(\tilde{\theta} \mid R_t)] + v_R \geq v_R + \mathbb{E}[1/\mathcal{I}(R_t)] = V(m)$, with equality achieved by $\widehat{R}_t$. For one-step forecasting, $y = C_{t+1}/m$. By H4, R_{t+1} is independent of $\mathcal{F}_t$, so $\mathbb{E}[(\hat{y} - y)^2 \mid \mathcal{F}_t] = (\hat{y} - \mathbb{E}y)^2 + \mathrm{Var}(y) \geq \mathrm{Var}(y) = V(m)$, with equality attained at $\hat{y} = \bar{R}$. □

S1.3. Derivation of Remark 1 (Reporting completeness versus coverage)

Writing $V(n, \rho) = v_R + \bar{R}/(\rho n) + (\bar{R}^2 + v_R)/(kn)$:

$$\lim_{\rho \to 1} V(n, \rho) = v_R + \frac{\bar{R}}{n} + \frac{\bar{R}^2 + v_R}{kn} > v_R, \qquad \lim_{n \to \infty} V(n, \rho) = v_R. \tag{S6}$$

Evaluating log-marginal effects:

$$-\frac{\partial V}{\partial \log n} = \frac{\bar{R}}{\rho n} + \frac{\bar{R}^2 + v_R}{kn} = \frac{A}{m}, \qquad -\frac{\partial V}{\partial \log \rho} = \frac{\bar{R}}{\rho n} = \frac{\bar{R}}{m}. \tag{S7}$$

Their ratio is $A/\bar{R} = 1 + \rho(\bar{R}^2 + v_R)/(k\bar{R})$.

S1.4. Proof of Proposition 5 (Multi-period allocation and cost optimization)

(i) Under H1–H4 with equal ancestor volume n per period, conditional on $(R_1, \ldots, R_T)$, counts across periods are independent. Letting $\bar{R}_T = T^{-1} \sum_{t=1}^{T} R_t$, by the law of total variance:

$$\frac{1}{T^2} \mathrm{Var}\Big(\sum_{t=1}^{T} \widehat{R}_t \Big) = \mathrm{Var}(\bar{R}_T) + \frac{1}{T^2} \sum_{t=1}^{T} \mathbb{E}\left[\frac{R_t}{\rho n} + \frac{R_t^2}{kn} \right] = \mathrm{Var}(\bar{R}_T) + \frac{A}{Tm}. \tag{S8}$$

The second term equals A/B, depending solely on total sample size $B = Tm$. When R_t is i.i.d., $\mathrm{Var}(\bar{R}_T) = v_R/T$, yielding Eq. (15). Under a stationary AR(1) process with $\mathrm{Cov}(R_s, R_t) = v_R\phi^{|s-t|}$, we have

$$T^2\mathrm{Var}(\bar{R}_T) = v_R T + 2v_R \sum_{h=1}^{T-1}(T-h)\phi^h = v_R\left[\frac{T(1+\phi)}{1-\phi} - \frac{2\phi(1-\phi^T)}{(1-\phi)^2}\right], \tag{S9}$$

which can be verified by induction on T. Strict monotonicity in T follows from the concavity of partial sums of geometric series.

(ii) Substituting $T = \mathcal{B}/(c_0 + c_1 m)$ into Eq. (15):

$$f(m) = \frac{(c_0 + c_1 m)(v_R + A/m)}{\mathcal{B}} = \frac{c_0 v_R + c_1 A + c_1 v_R m + c_0 A/m}{\mathcal{B}}. \tag{S10}$$

Differentiating yields $f'(m) = (c_1 v_R - c_0 A/m^2)/\mathcal{B}$ and $f''(m) = 2c_0 A/(m^3\mathcal{B}) > 0$. For $c_0 > 0$, f is strictly convex on $(0, \infty)$, with unique stationary point $m^{*2} = c_0 A/(c_1 v_R) = (c_0/c_1)m_\times$. Substituting m^* yields Eq. (17). □

S1.5. Proof of Outbreak Detection Test Properties

For $R > R'$, the likelihood ratio from Eq. (3) is

$$\frac{f(c; R)}{f(c; R')} = \frac{(k + \rho R')^{nk}\,[R(k + \rho R')]^c}{(k + \rho R)^{nk}\,[R'(k + \rho R)]^c}. \tag{S11}$$

Because $R/(k + \rho R)$ is strictly increasing in R, the base $R(k + \rho R')/[R'(k + \rho R)] > 1$. Thus, the likelihood ratio is strictly increasing in c, establishing that the family possesses a monotone likelihood ratio (MLR) in C_t.

(a) **Size:** By standard MLR theory, the power function of any test of the form $\{C_t \geq c\}$ is non-decreasing in R_t (Lehmann and Romano, 2005). Hence, the maximum rejection probability on H_0: $R_t \leq 1$ is attained at $R_t = 1$, where it equals $\alpha^* = P_{R_t=1}(C_t \geq c_\alpha) \leq \alpha$.

(b) **Uniform optimality:** By the Karlin–Rubin theorem (Karlin and Rubin, 1956), the test rejecting when $C_t \geq c_\alpha$ is uniformly most powerful (UMP) at level α^*.

(c) **Power:** Under $R_t = 1 + \delta$, $C_t \sim \mathrm{NB}(nk, k/[k + \rho(1 + \delta)])$, yielding Eq. (18). □

S1.6. Proof of Proposition 3 and Corollary 1 (Reporting Heterogeneity)

(i) Given (n, R_t, p_t), $\mathbb{E}(C_t \mid \cdot) = p_t n R_t$ and $\mathrm{Var}(C_t \mid \cdot) = p_t n R_t + p_t^2 n R_t^2/k$. Applying the law of total variance over p_t, and using $\mathbb{E}p_t = \rho$ and $\mathbb{E}p_t^2 = \rho^2 + \sigma^2$:

$$\mathrm{Var}(C_t \mid n, R_t) = \rho n R_t + \frac{(\rho^2 + \sigma^2) n R_t^2}{k} + \sigma^2 n^2 R_t^2. \tag{S12}$$

(ii) Unbiasedness follows from $\mathbb{E}(C_t \mid n, R_t) = \rho n R_t$, so $\mathbb{E}\widehat{R}_t = \bar{R}$. Applying total variance over R_t and dividing by $(\rho n)^2$:

$$V_H(m) = v_R + \frac{\sigma^2}{\rho^2}(\bar{R}^2 + v_R) + \frac{\bar{R}}{m} + \frac{(\rho^2 + \sigma^2)(\bar{R}^2 + v_R)}{\rho k m}, \tag{S13}$$

which yields Eq. (11) with $v_H = \tau^2(\bar{R}^2 + v_R)$.

(iii) Let $\theta_t = p_t R_t = \rho R_t u_t$. The Markov chain $R_t \to \theta_t \to C_t$ and the data processing inequality for Fisher information (Zamir, 1998) yield $\mathcal{I}_{C_t}(R_t) \le \mathcal{I}_{\theta_t}(R_t) = \mathcal{J}_u / R_t^2$, where $\mathcal{J}_u$ is the scale information of u_t. By Cauchy–Schwarz, $\mathcal{J}_u \ge 1/\tau^2$, with equality attained if and only if u_t follows a Gamma distribution with shape and rate $1/\tau^2$.

(iv) Unidentifiability: Note that $v_R + v_H = U - \bar{R}^2$ and $A_H = \bar{R} + \rho U/k$, where $U = (1+\tau^2)(\bar{R}^2 + v_R)$. The marginal distribution of $(C_1, \ldots, C_T)$ up to second moments depends on (v_R, τ^2) exclusively through the scalar combination U.

(v) Denominator lag signature: Let $D_t = \sum_s w_s C_{t-s}$. Under AR(1) reporting shock $u_t = 1 + \epsilon_t$ with persistence ψ, first-order delta expansion yields Eq. (13). Evaluating $F(w, \psi)/F(w', \psi)$ across $\psi \in [0, 1)$ yields the analytical bounds $q_1 \in [0.883, 1.338]$ and $q_2 \in [0.892, 1.374]$. □

For Corollary 1, differentiating V_H with respect to $\log \rho$ and $\log n$:

$$-\rho \frac{\partial V_H}{\partial \rho} - \left(-n \frac{\partial V_H}{\partial n}\right) = \frac{\bar{R}^2 + v_R}{kn} \left[-\rho \frac{\mathrm{d}\tau^2}{\mathrm{d}\rho}(kn + 1) - (1 + \tau^2)\right]. \tag{S14}$$

Under institutional reporting ($\mathrm{d}\tau^2/\mathrm{d}\rho = 0$), the difference is strictly negative. Under binomial reporting ($\tau^2 = (1-\rho)/[\rho(\nu+1)]$), setting the bracket positive yields $kn > \rho\nu \iff m > \rho^2 \nu / k$. □

S1.7. Proof of Proposition 6 (Precision of Environmental Variance Estimation)

Supplementary Proposition S1 (Finite-period precision and Fisher information for environmental variance)**.** *Under H1–H4 and H6, with equal sample volume m per period:*

(a) When R_t is i.i.d., the modified moment estimator $\hat{v}_R$ in Eq. (19) has bias $-\rho\sigma_Y^2/[kmT(1+\rho/(km))]$ and variance

$$\mathrm{Var}(\hat{v}_R) = \frac{1}{(1+\rho/(km))^2}\left[\frac{\mu_{4Y}}{T} - \frac{(T-3)\sigma_Y^4}{T(T-1)} - \frac{2\varpi'\mu_{3Y}}{T} + \frac{\varpi'^2\sigma_Y^2}{T}\right]\left(1+O\left(\frac{1}{mT}\right)\right), \tag{S15}$$

where $\varpi' = (1+2\rho\bar{R}/k)/m$. As $m\to\infty$, $T\mathrm{Var}(\hat{v}_R)\to\Omega_\infty = \mathbb{E}(R_t-\bar{R})^4 - v_R^2$. When $R_t \sim$ Gamma(ξ,ϑ), $\Omega_\infty = v_R^2(2+6v_R/\bar{R}^2)$.

(b) Let the joint distribution of $(R_1,\ldots,R_T)$ belong to a family parameterized by v_R. For any scale m and any correlation structure, $\mathcal{I}_{C_{1:T}}(v_R) \le \mathcal{I}_{R_{1:T}}(v_R)$. When periods are independent with $R_t \sim$ Gamma of fixed mean $\bar{R}$:

$$\mathcal{I}_{R_{1:T}}(v_R) = \frac{T\xi\,[\xi(\ln\Gamma)''(\xi)-1]}{v_R^2}, \qquad \xi = \frac{\bar{R}^2}{v_R}, \tag{S16}$$

where $(\ln\Gamma)''$ is the trigamma function. For stationary Gaussian AR(1) environments, $\mathcal{I}_{R_{1:T}}(v_R) = T/(2v_R^2)$, independent of ϕ.

Proof. Item (i) in Proposition 5 follows from setting sample variance S_C^2 equal to marginal variance $m^2V(m) = m^2v_R + m\bar{R} + (\rho m/k)(\bar{R}^2+v_R)$ and solving for v_R.

For Supplementary Proposition S1(a), standard moments of sample variance give Eq. (S15). As $m\to\infty$, $Y_t\to R_t$, recovering $\Omega_\infty = \mathrm{Var}((R_t-\bar{R})^2) = \mathbb{E}(R_t-\bar{R})^4 - v_R^2$. For Gamma$(\xi,\vartheta)$, the fourth central moment is $3\xi(\xi+2)\vartheta^4$, which upon substituting $\xi = \bar{R}^2/v_R$ and $\vartheta = v_R/\bar{R}$ yields $\Omega_\infty = v_R^2(2+6v_R/\bar{R}^2)$.

For Supplementary Proposition S1(b), the Markov chain $v_R \to R_{1:T} \to C_{1:T}$ and the data processing inequality yield $\mathcal{I}_{C_{1:T}}(v_R) \le \mathcal{I}_{R_{1:T}}(v_R)$. Evaluating the Fisher information matrix for the two-parameter Gamma family under the chain rule $\mathrm{d}\xi/\mathrm{d}v_R = -\bar{R}^2/v_R^2$ and $\mathrm{d}\vartheta/\mathrm{d}v_R = 1/\bar{R}$ yields Eq. (S16). Under stationary Gaussian AR(1), $R_{1:T}-\bar{R} \sim \mathcal{N}(0, v_R\Sigma_\phi)$, giving Fisher information $\frac{1}{2}\mathrm{tr}\{[(v_R\Sigma_\phi)^{-1}\Sigma_\phi]^2\} = T/(2v_R^2)$.

Finally, for Gaussian AR(1), $\mathrm{Cov}[(R_s-\bar{R})^2,(R_t-\bar{R})^2] = 2v_R^2\phi^{2|s-t|}$. Summing over pairs yields $2v_R^2\kappa_T(\phi^2)/T$, confirming Eq. (20). □

S1.8. Proof of Corollary 2 (Season-Level Variance Bound)

Forming the Markov chain $v_S \to (\eta_1, \ldots, \eta_S) \to (R_t) \to (C_t)$, the data processing inequality establishes that the total Fisher information for v_S cannot exceed the information in the latent seasonal shocks: $\mathcal{I}(v_S) \le S/(2v_S^2)$. By Cramér–Rao, any unbiased estimator satisfies $\mathrm{Var}(\hat{v}_S) \ge 2v_S^2/S$, proving the $\sqrt{2/S}$ relative error bound. When seasonal means are unknown, sample variance degrees of freedom reduce to $S-1$. □

S1.9. Proof of Proposition 7 (Budget Threshold under AR(1) Environments)

Substituting the budget constraint into $f_\phi(T) = v_R e(T) + Ac_1/(\mathcal{B} - c_0 T)$, the first-order condition requires $v_R g_T = c_0 A/(c_1 m^2)$. Monotonicity of $g_T(\phi)$ from $g_1 < 1$ to $(1+\phi)/(1-\phi) > 1$ guarantees a unique root $T^\dagger$ where $g_{T^\dagger} = 1$, corresponding to budget $\mathcal{B}^\dagger = T^\dagger(c_0 + c_1 m_0^*)$. When $\mathcal{B} < \mathcal{B}^\dagger$, $T^* < T^\dagger$ and $g_{T^*} < 1$, requiring $m^* > m_0^*$. When $\mathcal{B} > \mathcal{B}^\dagger$, $m^* < m_0^*$. □

S1.10. Shrinkage and Empirical Bayes Estimation

Consider the linear shrinkage estimator $\hat{\theta}_\lambda = (1-\lambda)C_t/m + \lambda\mu_0$, where μ_0 is a prior benchmark and $\delta = \mu_0 - \bar{R}$. The mean squared error is

$$\mathrm{MSE}(\lambda) = (1-\lambda)^2 V + \lambda^2 \delta^2, \qquad V = v_R + \frac{A}{m}. \tag{S17}$$

Minimizing over λ yields $\lambda^* = V/(V+\delta^2)$ and $\mathrm{MSE}(\lambda^*) = V\delta^2/(V+\delta^2)$. Thus, $\mathrm{MSE}(\lambda^*) < v_R$ if and only if $\delta^2 < v_R(1 + m/m_\times)$. The estimator achieves variance reduction by introducing conditional bias $\lambda(\mu_0 - R_t)$. □

S1.11. Renewal Process Variance and Delta Expansion of Denominator Noise

Conditional on history $\mathcal{F}_t$, let $\bar{\Lambda} = \mathbb{E}(\Lambda_{t+1} \mid \mathcal{F}_t)$ and $m = \rho\bar{\Lambda}$. By the law of total variance conditioned on (Λ_{t+1}, R_{t+1}):

$$\begin{aligned}\mathrm{Var}(C_{t+1} \mid \mathcal{F}_t) &= \mathbb{E}\left[\rho\Lambda_{t+1}R_{t+1} + \frac{\rho^2\Lambda_{t+1}R_{t+1}^2}{k} \;\middle|\; \mathcal{F}_t\right] + \mathrm{Var}(\rho\Lambda_{t+1}R_{t+1} \mid \mathcal{F}_t) \\ &= mA + \rho^2\bar{\Lambda}^2\left[v_R + \mathrm{CV}^2(\Lambda_{t+1} \mid \mathcal{F}_t)\,(\bar{R}^2 + v_R)\right]. \end{aligned} \tag{S18}$$

Dividing by $m^2 = (\rho\bar{\Lambda})^2$ yields Eq. (23).

For the denominator noise expansion, condition on true infection history $\mathcal{H}_t$, and define $D_t = m(1+\delta)$ where $m = \rho\Lambda_{t+1}$ and $c_w = (1-\rho)\sum_s w_s^2$.

Then $\mathbb{E}(\delta) = 0$ and $\mathbb{E}(\delta^2) = c_w/m$. Expanding $G_t = (C_{t+1}/m)(1+\delta)^{-1}$ to second order in δ:

$$\mathbb{E}(G_t \mid \mathcal{H}_t) = \bar{R}\left(1 + \frac{c_w}{m}\right) + O(m^{-2}), \qquad \mathrm{Var}(G_t \mid \mathcal{H}_t) = v_R + \frac{A + (\bar{R}^2 + 3v_R)c_w}{m} + O(m^{-2}). \tag{S19}$$

Normalizing by the squared conditional expectation $[\mathbb{E}(G_t \mid \mathcal{H}_t)]^2 = \bar{R}^2[1 + 2c_w/m + O(m^{-2})]$ eliminates $2v_R c_w/m$, yielding the slope expression in Eq. (25). □

S2. Robustness analyses and supplementary tables

This section documents the robustness checks, sensitivity specifications, and supplementary tables referenced in the main text.

S2.1. Equivalence testing of reporting adjustment

Table S3 presents the two one-sided tests for equivalence (TOST) evaluating the impact of reporting completeness adjustment $\Delta a = a - a^*$ across seven pathogen-window combinations. All three influenza combinations demonstrate statistical equivalence under both BCa bootstrap intervals and normal approximation tests at the 5% margin (± 0.0050). Calendar windows for COVID-19 and RSV achieve equivalence under the BCa interval criterion. The two growth-phase windows exceed equivalence margins primarily because their baseline intercepts are small ($a \approx 0.02$–0.03), making the 5% equivalence band extremely narrow (± 0.001). For the 2025–2026 influenza season, $\Delta a = +0.0039$ (90% BCa CI 0.0020–0.0060), which remains within the seasonal equivalence margin of 0.0088.

S2.2. Evaluation of reporting detection power on simulated grids

Table S4 evaluates the sensitivity of the Δa detection metric across a 35-cell simulation grid of reporting volatility. When reporting fluctuations are uncorrelated with hospital reporting completeness ($\kappa = 0$), the false rejection rate is 0 across all cells. As elasticity κ increases, power to detect reporting effects rises; however, because empirical reporting completeness is concentrated near unity (median 0.943), observable institutional completeness explains $< 1\%$ of intercept elevation, while within-facility underreporting or clinical threshold shifts ($\tau > 0$) remain unconstrained by Δa.

S2.3. Epidemic season random-effects pooling

Table S5 reports season-specific intercept estimates and random-effects pooling using the DerSimonian–Laird method. Individual season intercepts range from 0.028 to 0.135 ($Q = 47.4, p = 5 \times 10^{-11}$). The pooled intercept across the three primary seasons is $a_{\mathrm{RE}} = 0.077$ (95% CI -0.056 to 0.211; $\hat{v}_S = 0.0028$), illustrating the theoretical precision lower bound $\sqrt{2/(S-1)} = 1.00$ for $S = 3$.

S2.4. Generation interval aggregation sensitivity

Table S6 evaluates sensitivity to discrete generation block length (7 days versus 4 days). Decreasing block length scales baseline intercept magnitude down to approximately 60% under independent environments, but intercept ratios q_1, q_2 vary by less than 0.05. Intercept ratios are driven primarily by admission delay convolution rather than generation block aggregation.

S2.5. Independent non-NHSN surveillance streams

Table S7 presents scaling estimates from outpatient ILINet ($N = 1{,}479$) and hospital FluSurv-NET ($N = 364$). ILINet reproduces the directional intercept pattern ($q_1 = 0.687, q_2 = 1.467$, with 99.7% of bootstrap resamples satisfying $q_1 < 1 < q_2$). FluSurv-NET covers only 14 sentinel sites, producing wide confidence intervals that span unity.

S2.6. Null range specifications and evidence grading

Table S8 details pre-specified null simulation ranges and sensitivity classifications across the seven pathogen-window combinations. Under pre-specified full-grid criteria, 4 combinations achieve partial replication and 3 show directional consistency (0/4/3). Under sensitivity specifications excluding admission delays, full replication is achieved in 2 combinations.

S2.7. Supplementary figures

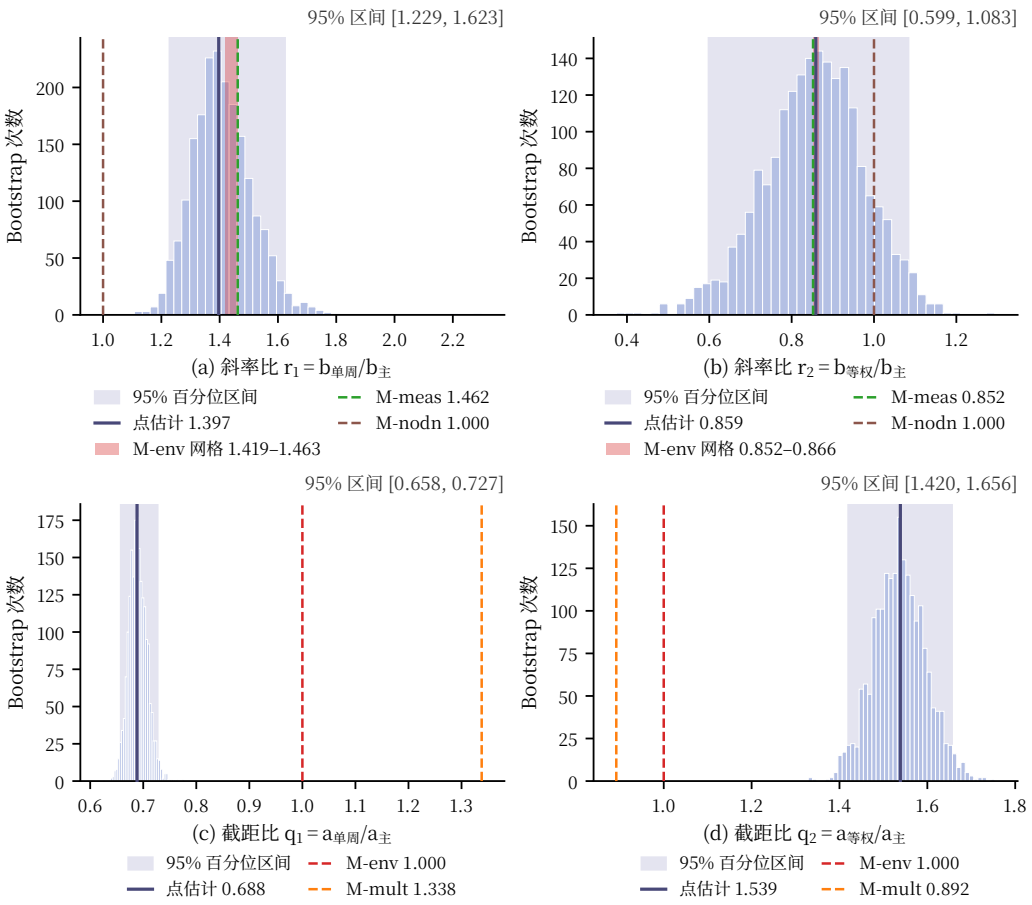


Figure S1: State-clustered bootstrap distributions of lag-weight ratios ($N = 1{,}267$ state-weeks, $B = 2{,}000$ resamples). (A) Slope ratio $r_1 = b_{\text{single}}/b_{\text{primary}}$. (B) Slope ratio $r_2 = b_{\text{equal}}/b_{\text{primary}}$. (C) Intercept ratio $q_1 = a_{\text{single}}/a_{\text{primary}}$. (D) Intercept ratio $q_2 = a_{\text{equal}}/a_{\text{primary}}$. Vertical solid lines indicate point estimates; grey shaded areas represent 95% percentile confidence intervals; tick markers indicate theoretical predictions from candidate mechanisms (Table S2). Slope ratios exclude the noise-free denominator model (M-nodn), while intercept ratios reject stationary models but overlap with dynamic growth renewal ranges.

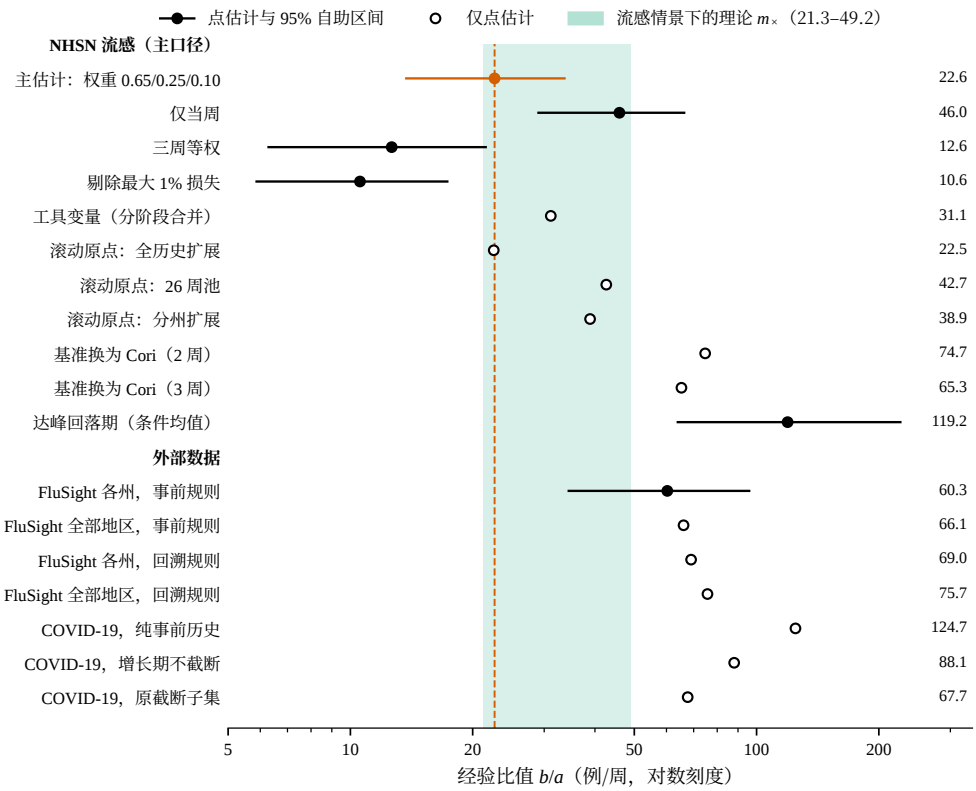


Figure S2: Forest plot of empirical balance scale ratios b/a across model specifications (cases/week, logarithmic scale). Horizontal bars represent 95% confidence intervals from cluster bootstrap resamples. Shaded band indicates the theoretical scenario range $m_\times \in [21.3, 49.2]$, with vertical dashed line marking the baseline estimate of 22.6 cases/week. Upper section displays NHSN influenza sensitivity specifications; lower section displays external surveillance streams.

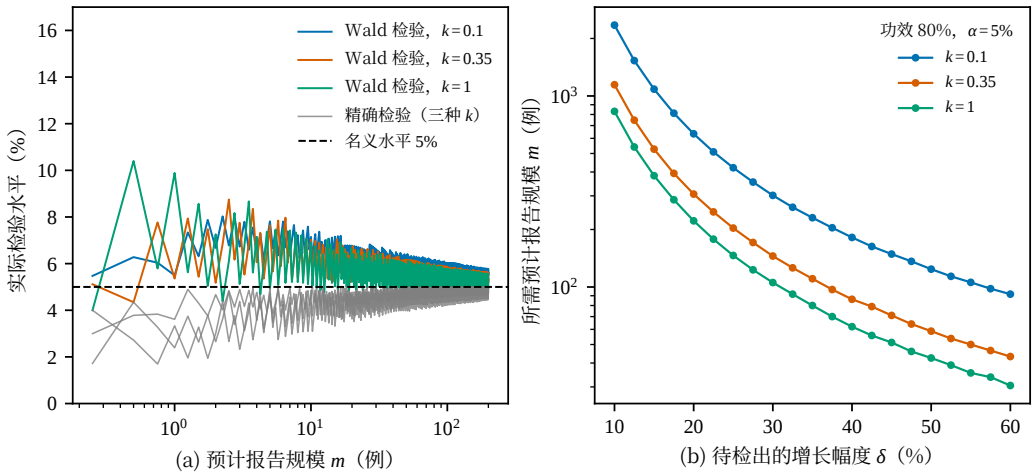


Figure S3: Exact test size and required sample volume for outbreak detection ($\rho = 0.25$, nominal size $\alpha = 0.05$, exact negative-binomial probabilities). (A) Actual test size of standard Wald tests (colored curves by dispersion k) versus exact uniformly most powerful tests (grey curve) as a function of expected reporting scale m. (B) Expected reporting volume required to achieve and maintain $\geq 80\%$ statistical power versus detectable transmission excess $\delta = R_t - 1$.

Table S1: Comparison of three functional mean shapes for estimation loss on the early-epidemic influenza sample ($N = 1{,}267$ state-weeks). Models are estimated by Gamma quasi-likelihood. Cross-validation is leave-one-state-out, with baseline mean $\bar{R}$ re-estimated solely on training states in each fold. Deviance differences are reported relative to the affine specification M1, with standard errors across states in parentheses.

Model	Parameter estimates	AIC	CV deviance	ΔDeviance vs M1	States favoring
M0: $L = b/D$	$b = 21.7$	−1971.2	5407.1	1949.9 (243.1)	48/51
M1: $L = a + b/D$	$a = 0.0991$, $b = 2.243$	−2718.8	3457.2	–	–
M2: $L = cD^{-p}$	$c = 0.512$, $p = 0.273$	−2692.6	3514.0	56.8 (27.2)	28/51

S3. Supplementary tables and robustness analyses

This section compiles the supplementary tables supporting the empirical analysis and robustness evaluations.

Table S2: Empirical confidence intervals for lag-weight ratios and theoretical predictions across noise mechanisms (NHSN early-phase influenza, $N = 1{,}267$ state-weeks). r_1, r_2 denote slope ratios for current-week and three-week equal weighting relative to primary weighting; q_1, q_2 denote corresponding intercept ratios; a is the baseline intercept. Empirical intervals are 95% percentile intervals from 2,000 state-clustered bootstrap replications. Stationary environmental models (M-env) predict $q = 1$. Renewal process simulations under growth and admission delays widen the null range. Multiplicative reporting shock models (M-mult) evaluate AR(1) reporting persistence $\psi \in [0, 1)$. M-bin0 is binomial observation without environmental variance; M-meas is pure measurement error; M-nodn assumes a noise-free denominator. Bold entries indicate theoretical ranges that do not intersect empirical confidence intervals.

Mechanism	r_1	r_2	q_1	q_2	a
Empirical	1.397	0.859	0.688	1.539	0.0991
95% percentile CI	1.229–1.623	0.599–1.083	0.658–0.727	1.420–1.656	0.0837–0.1165
M-env, stationary	1.419–1.463	0.852–0.866	**1**	**1**	> 0
M-env, growth phase, weekly	1.450–1.630	0.693–0.763	**0.857–0.937**	**1.226–1.333**	> 0
M-env, daily renewal, no delay	–	–	**0.746–0.887**	1.319–1.484	> 0
M-env, daily renewal, all delays	–	–	0.601–0.887	1.187–1.568	> 0
M-mult, stationary	1.462	0.852	**0.883–1.338**	**0.892–1.374**	> 0
M-mult, growth phase	–	–	0.686–1.224	0.865–1.497	> 0
M-bin0	1.432	0.862	No intercept	No intercept	**0**
M-meas	1.462	0.852	No intercept	No intercept	**0**
M-nodn	**1.000**	1.000	**1**	**1**	> 0

Table S3: Two one-sided tests for equivalence (TOST) on the observable reporting completeness contribution $\Delta a = a - a^*$ across seven pathogen-window combinations (2,000 state-clustered bootstrap resamples, 90% BCa intervals). The equivalence margin is ±5% of the baseline intercept a. Equivalence under BCa requires the 90% BCa interval to fall entirely within the margin; normal equivalence evaluates two one-sided Wald tests at level 0.05. Intercept excess is defined as $a - 0.08/\bar{R}^2$.

Combination	N	a	Excess	Δa	90% BCa CI	Margin	BCa equivalent	Normal equivalent
Influenza, calendar window	1,267	0.099	0.070	−0.0004	−0.0012–0.0005	0.0050	Yes	Yes
Influenza, growth window	2,506	0.102	0.070	−0.0011	−0.0037–0.0021	0.0051	Yes	Yes
Influenza, 2025–2026 season	663	0.177	0.145	0.0039	0.0020–0.0060	0.0088	Yes	Yes
COVID-19, calendar window	1,479	0.037	−0.031	0.0001	−0.0004–0.0009	0.0018	Yes	No
COVID-19, growth window	2,774	0.023	−0.030	−0.0000	−0.0064–0.0055	0.0012	No	No
RSV, 2024–2025 season	494	0.080	0.037	0.0010	−0.0006–0.0023	0.0040	Yes	No
RSV, growth window	1,011	0.028	−0.010	0.0036	−0.0011–0.0131	0.0014	No	No

Table S4: Test size and power of the Δa detection criterion across a simulation grid of reporting volatility (influenza early-phase calendar window, reporting probability $\rho h^{\kappa} u$, 35 cells, 200 replications per cell, each with $B = 200$ resamples). τ denotes reporting volatility independent of hospital completeness h; κ represents the elasticity of reporting probability to completeness h. When $\kappa = 0$, reporting is independent of completeness, and false rejection rate is zero across all cells.

	$\kappa = 0$	$\kappa = 1$	$\kappa = 3$	$\kappa = 5$	$\kappa = 8$
(a) Cell mean of intercept a					
$\tau = 0$	0.100	0.099	0.101	0.104	0.111
$\tau = 0.05$	0.104	0.104	0.105	0.109	0.115
$\tau = 0.1$	0.118	0.118	0.118	0.121	0.127
$\tau = 0.15$	0.140	0.138	0.140	0.144	0.149
$\tau = 0.2$	0.172	0.169	0.169	0.174	0.181
$\tau = 0.3$	0.257	0.255	0.251	0.255	0.265
$\tau = 0.5$	0.523	0.527	0.523	0.521	0.530
(b) Cell mean of difference Δa					
$\tau = 0$	−0.0016	−0.0004	0.0009	0.0018	0.0028
$\tau = 0.05$	−0.0018	−0.0004	0.0008	0.0019	0.0028
$\tau = 0.1$	−0.0018	−0.0004	0.0009	0.0019	0.0028
$\tau = 0.15$	−0.0019	−0.0006	0.0009	0.0019	0.0029
$\tau = 0.2$	−0.0024	−0.0008	0.0009	0.0018	0.0030
$\tau = 0.3$	−0.0031	−0.0010	0.0010	0.0017	0.0031
$\tau = 0.5$	−0.0060	−0.0032	0.0001	0.0021	0.0033
(c) Proportion of replications with lower BCa bound of $\Delta a > 0$					
$\tau = 0$	0.000	0.010	0.465	0.840	0.945
$\tau = 0.05$	0.000	0.005	0.405	0.860	0.945
$\tau = 0.1$	0.000	0.025	0.415	0.820	0.920
$\tau = 0.15$	0.005	0.005	0.345	0.740	0.920
$\tau = 0.2$	0.000	0.010	0.240	0.635	0.850
$\tau = 0.3$	0.000	0.020	0.220	0.465	0.735
$\tau = 0.5$	0.000	0.020	0.055	0.225	0.445
(d) Median ratio of upper BCa bound of Δa to intercept excess					
$\tau = 0$	0.0000	0.0118	0.0301	0.0460	0.0626
$\tau = 0.05$	0.0000	0.0107	0.0280	0.0441	0.0571
$\tau = 0.1$	0.0000	0.0092	0.0262	0.0375	0.0506
$\tau = 0.15$	0.0000	0.0092	0.0225	0.0334	0.0423
$\tau = 0.2$	0.0000	0.0082	0.0185	0.0255	0.0357
$\tau = 0.3$	0.0000	0.0065	0.0154	0.0175	0.0251
$\tau = 0.5$	0.0000	0.0054	0.0098	0.0126	0.0160

Table S5: Season-specific intercepts and random-effects pooling (NHSN influenza, 95% percentile intervals from 2,000 state-clustered bootstrap resamples; pooled via DerSimonian–Laird estimator with t_{S-1} intervals; between-season variance $\hat{v}_S$ interval via Q-profile method). (a) Individual season estimates, with SE denoting bootstrap standard error across states. (b) Pooled results; the final column provides theoretical relative standard error bounds $\sqrt{2/S}$ and $\sqrt{2/(S-1)}$ from Corollary 2.

(a) Epidemic season	N	a	95% CI	SE
2022–2023 season (calendar window)	338	0.071	0.049–0.094	0.0121
2023–2024 season (calendar window)	419	0.028	0.017–0.041	0.0077
2024–2025 season (calendar window)	510	0.135	0.111–0.166	0.0139
2025–2026 season (growth window, out-of-sample)	663	0.177	0.141–0.212	0.0186

(b) Pooled	a_{RE}	95% t CI	$\hat{v}_S$	Q-profile CI	Q	I^2	Lower bound
$S = 3$	0.077	−0.056–0.211	0.0028	0.0007–0.1155	47.4	0.958	0.82 / 1.00
$S = 4$	0.101	−0.002–0.205	0.0041	0.0012–0.0609	85.0	0.965	0.71 / 0.82

Table S6: Simulated intercepts and intercept ratios under 7-day and 4-day generation block aggregations (influenza early-phase calendar window, intermediate generation interval, environmental squared coefficient of variation $\mathrm{CV}^2 = 0.0287$; 20 of 60 simulation cells, 300 replications per cell). ϕ denotes weekly environmental autocorrelation, mapped to $\phi^{4/7}$ for 4-day blocks. Asterisks denote cells falling simultaneously within the empirical 95% percentile CI of a and 95% BCa CIs of q_1, q_2. Across all 30 cell pairs, the maximum absolute difference between block lengths is 0.088 for a, 0.029 for q_1, and 0.049 for q_2. Empirical values are $a = 0.099$, $q_1 = 0.688$, $q_2 = 1.539$, and $b/a = 22.6$.

ϕ	Delay	a (7d)	a (4d)	q_1 (7d)	q_1 (4d)	q_2 (7d)	q_2 (4d)	b/a (7d)	b/a (4d)
0	None	0.100	0.056	0.824	0.816	1.381	1.398	14.2	25.5
0	Long	0.073	0.044	0.722	0.727	1.528	1.518	18.6	30.7
0.3	None	0.126	0.109*	0.735	0.712	1.512	1.546	10.6	12.2
0.3	Long	0.102*	0.094*	0.680	0.677	1.603	1.607	12.7	14.0
0.5	None	0.143	0.132	0.693	0.675	1.584	1.611	8.2	9.3
0.5	Long	0.123	0.121	0.658	0.657	1.645	1.639	9.7	9.7
0.7	None	0.157	0.150	0.665	0.655	1.645	1.657	6.0	6.7
0.7	Long	0.144	0.142	0.646	0.645	1.679	1.679	6.8	7.0
0.9	None	0.149	0.148	0.650	0.648	1.705	1.709	5.2	5.1
0.9	Long	0.147	0.147	0.649	0.646	1.712	1.715	5.0	5.0

Table S7: Intercept ratios in non-NHSN surveillance data (2022–2025 calendar windows, estimated via the primary pipeline; 95% percentile intervals from 2,000 cluster bootstrap resamples). The final column reports the proportion of resamples satisfying $q_1 < 1 < q_2$. HHS Protect daily hospital data correlate at $r = 0.995$ with NHSN across 3,417 state-weeks (median relative difference 0.074). Due to distinct case definitions, these datasets serve as directional consistency checks rather than formal replications.

Data stream	Clustering unit	N	q_1	q_2	Directional consistency
ILINet outpatient ILI	51 jurisdictions	1,479	0.687 (0.552–0.842)	1.467 (1.406–1.531)	0.997
FluSurv-NET admission rate	14 sites	364	0.686 (−3.608–3.920)	1.312 (−4.382–7.217)	0.560

Table S8: Null range specifications and evidence grading across seven pathogen-window combinations (95% BCa intervals from 2,000 state-clustered bootstrap resamples). The pre-specified null range spans the minimum to maximum across 108 daily renewal simulation cells and 48 multiplicative reporting shock cells. Classification columns: Primary (pre-specified full grid); No delay (daily renewal without admission delays); Weekly generation (weekly-generation renewal simulations); Median and Ratio of means denote post-hoc sensitivity specifications. Classifications: Replicated, Partial (partial replication), and Directional (directionally consistent).

Combination	N	q_1 CI	q_2 CI	q_1 Null range	q_2 Null range	Excess CI	Primary	No delay	Weekly	Median	Mean ratio
Influenza, calendar window	1,267	0.656–0.727	1.422–1.655	0.601–1.224	0.865–1.568	0.0530–0.0867	Partial	Partial	Replicated	Partial	Partial
Influenza, growth window	2,506	0.487–0.643	1.686–2.261	0.625–1.242	0.863–1.578	0.0455–0.0945	Partial	Replicated	Replicated	Replicated	Replicated
Influenza, 2025–2026 season	663	0.575–0.715	1.430–1.657	0.639–1.240	0.864–1.569	0.1100–0.1850	Partial	Partial	Replicated	Partial	Partial
COVID-19, calendar window	1,479	0.652–0.799	1.393–1.676	0.558–1.279	0.893–1.763	−0.0415—−0.0112	Directional	Directional	Partial	Directional	Directional
COVID-19, growth window	2,774	0.688–0.988	1.174–1.642	0.610–1.272	0.885–1.653	−0.0400—−0.0154	Directional	Directional	Directional	Directional	Directional
RSV, 2024–2025 season	494	0.618–0.774	1.381–1.799	0.210–3.679	−1.888–1.609	0.0245–0.0509	Partial	Replicated	Replicated	Partial	Partial
RSV, growth window	1,011	0.520–0.803	1.281–2.061	0.539–1.266	0.861–1.547	−0.0187–0.0057	Directional	Directional	Directional	Directional	Directional
Tally (Replicated/Partial/Directional)							0/4/3	2/2/3	4/1/2	1/3/3	1/3/3

S4. Replication package overview

The complete replication archive containing analytical code, data snapshots, and numerical result dictionaries is permanently maintained at https://github.com/wangchi2068/case-count-bottleneck.

Archive structure.

- `experiment/code/`: All 23 execution scripts for data processing, FGLS fitting, bootstrap resampling, and renewal simulations.
- `experiment/data/`: Processed panel datasets, including NHSN weekly hospitalisations, ILINet, and FluSurv-NET data.
- `experiment/analysis/`: Result JSON dictionaries and numerical mapping tables (`reproduction_map.json`, `reproduction_map.md`), linking every reported number in the manuscript to its programmatic output.
- `experiment/logs/`: Execution logs and figure quality audit records.

Computational environment. Python 3.11 with NumPy 2.4, SciPy 1.17, Pandas 3.0, and Matplotlib 3.11. All scripts execute locally without external API dependencies. Baseline reproducibility was verified across 24,815 numerical comparisons within 10^{-9} numerical tolerance.

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